

# System Topology & Flow Diagram (English)

## — v3.0

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Six perspectives from shallow to deep, static to dynamic — making LLM-ERP understandable to both technical and non-technical readers.

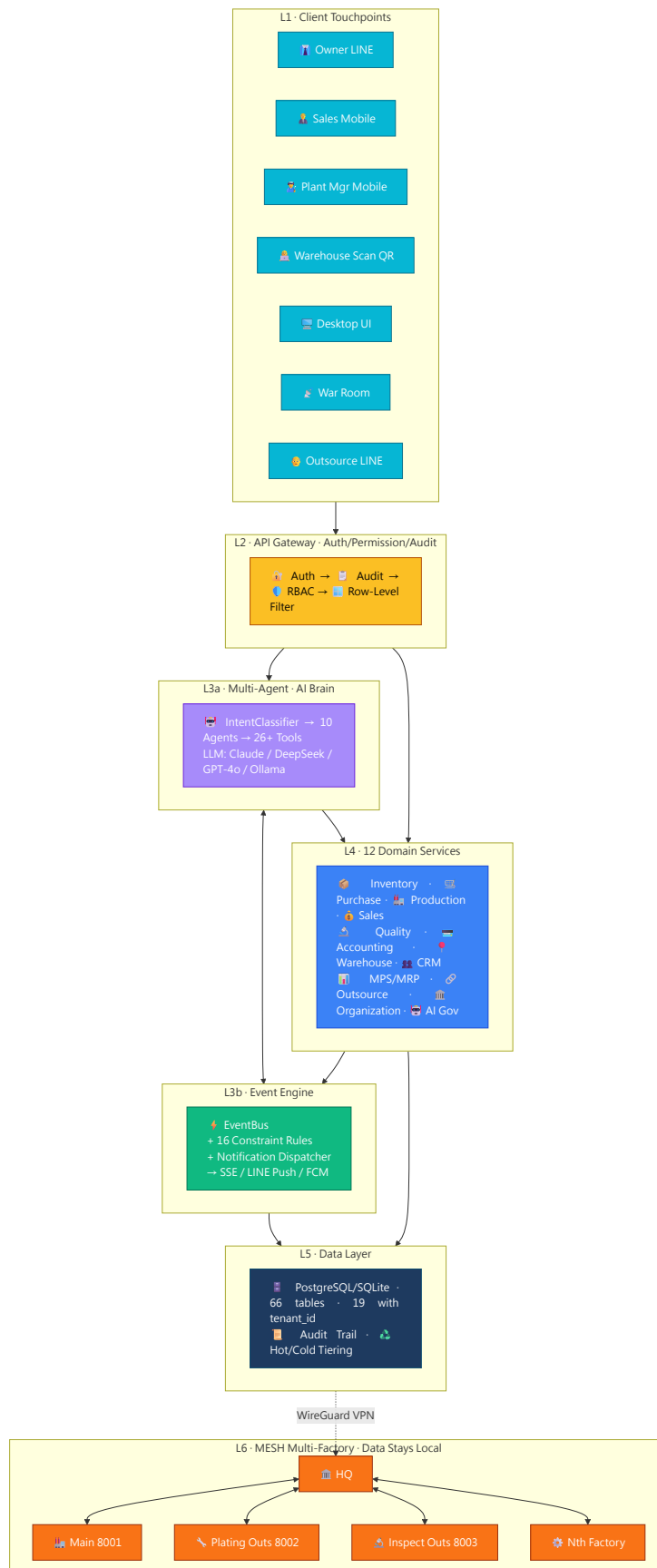
- **Visual:** [\\_system\\_flow\\_topology.svg\\_](#) (beautiful SVG for slides / A3 print)
- **Static legacy:** [\\_architecture\\_diagram.svg\\_](#) (5-layer)

Chinese version: [\\_SYSTEM\\_TOPOLOGY\\_ZH.md\\_](#)

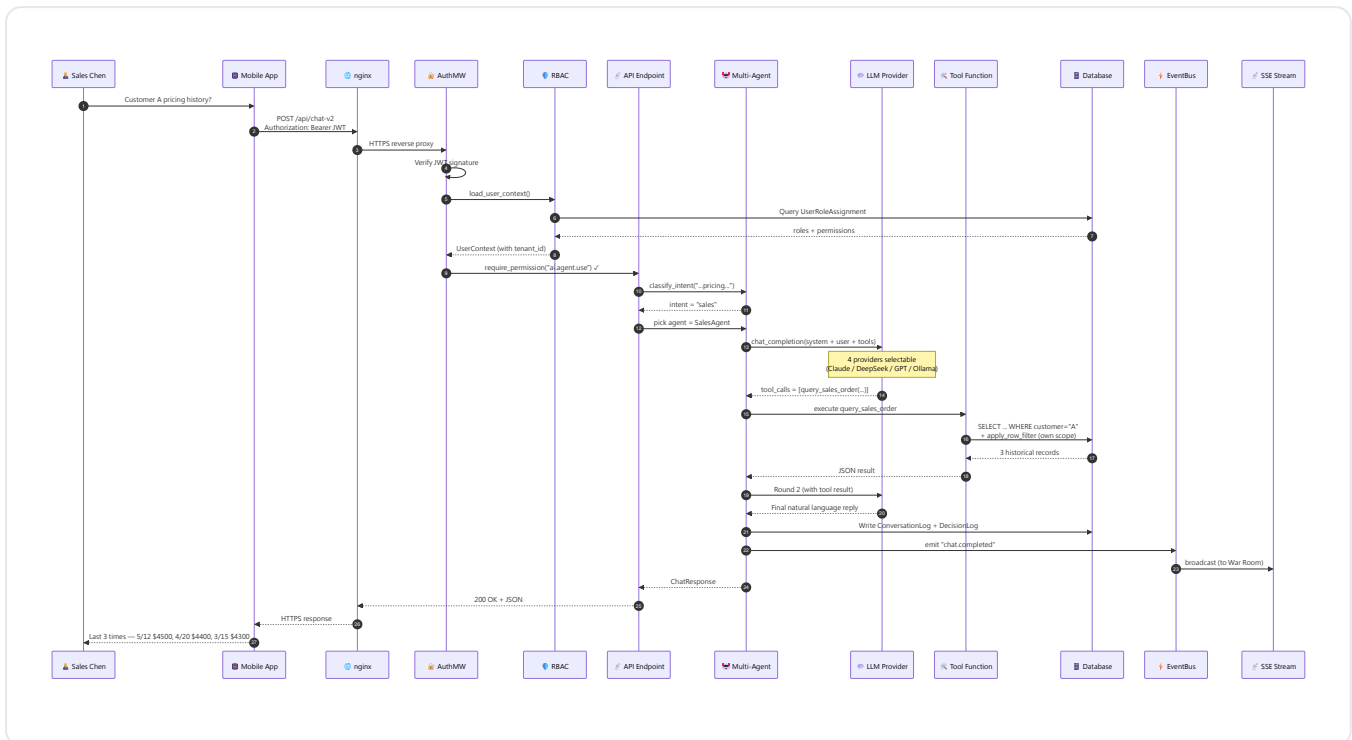
⚡ **v3.0 Strategic Pivot Notice:** SVG may still show "🤖 Outsource LINE / Plating Outsource / Inspection Outsource" nodes. v3.0 deprecates outsource / LINE / Mobile (moved to Phase 7). SVG redraw pending Phase 1 completion.

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## View 1: Six-Layer Overall Architecture (Owner 30-sec scan)



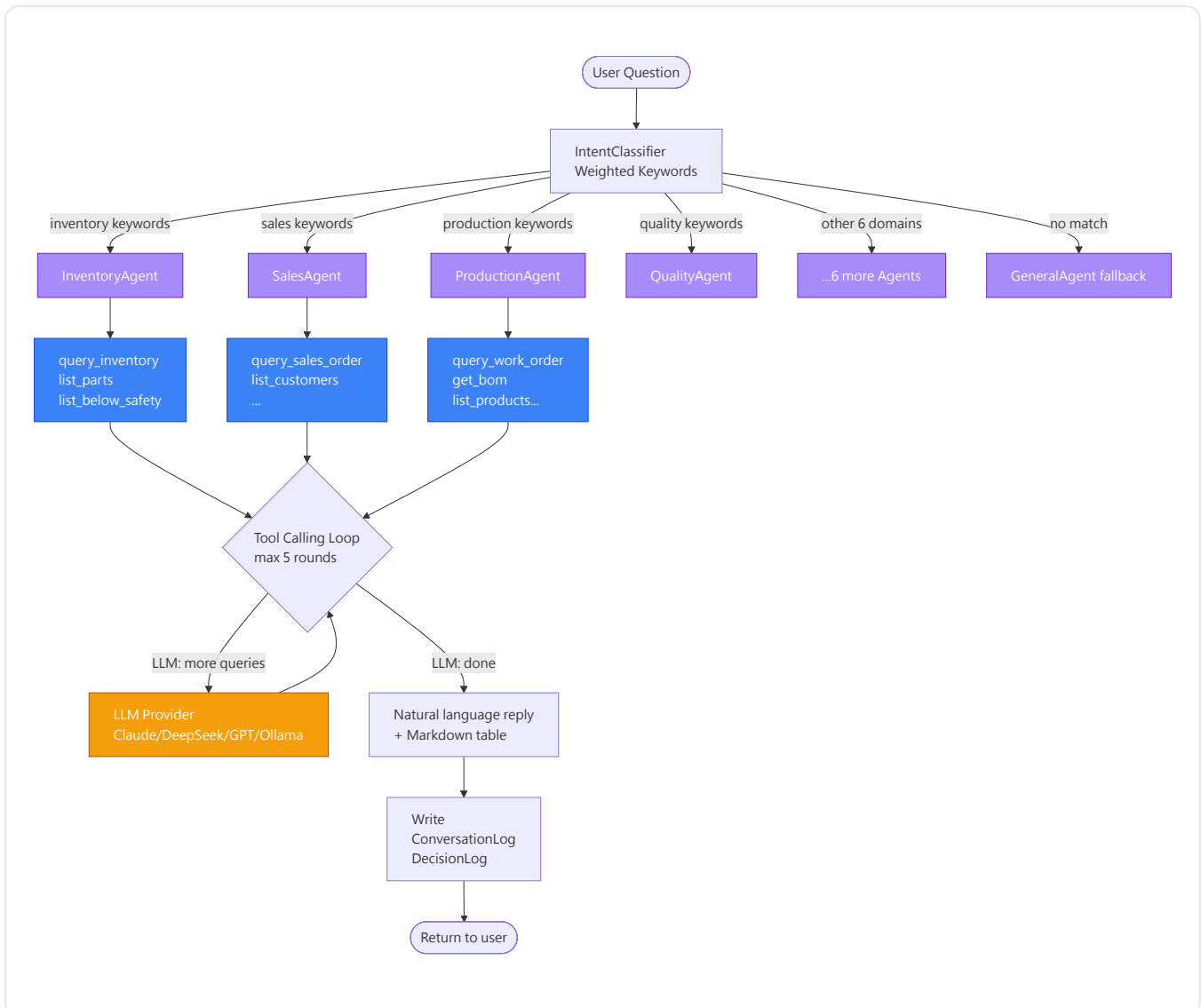
## View 2: Full Request Lifecycle (Engineer "how data flows")



### Key flow notes:

1. Steps 6-7: Permission load = single JOIN query, 5-min TTL cache
2. Step 11: IntentClassifier uses weighted keywords
3. Step 14: LLM tool calling can loop up to 5 rounds
4. Step 18: `apply_row_filter` auto-adds `WHERE created_by = chen` (Chen sees only own customers)
5. Step 23: All AI decisions written to DecisionLog for audit

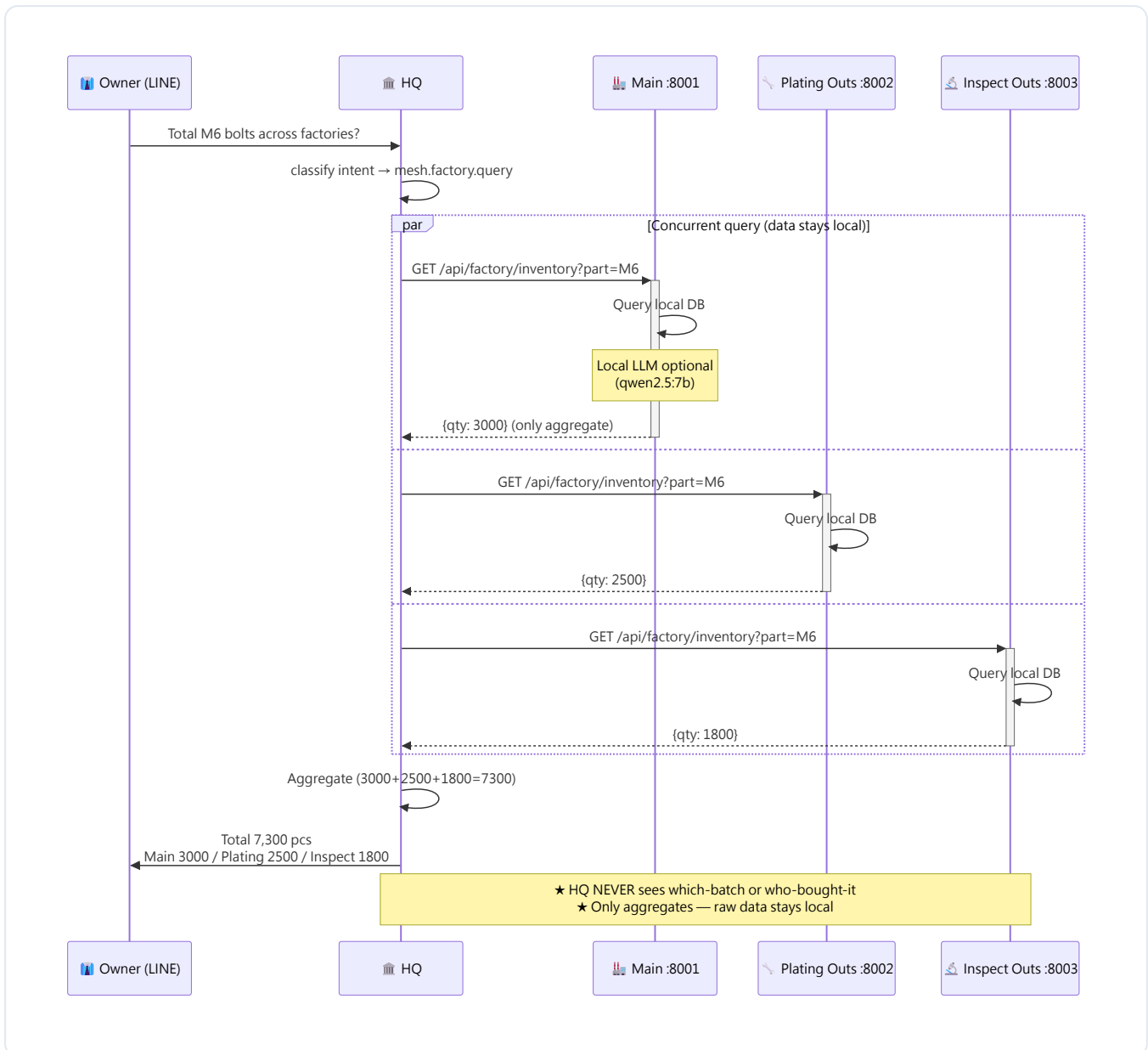
## View 3: Multi-Agent Internals (AI Engineer)



Each Agent has scoped tools — preventing LLM from misfiring cross-domain:

- **InventoryAgent** only sees 4 inventory tools
- **SalesAgent** only sees 4 sales tools
- → Inventory agent cannot accidentally trigger sales operations

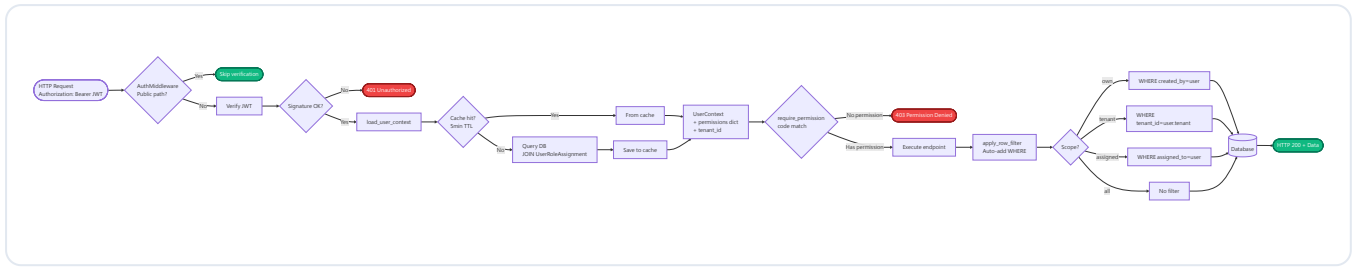
## View 4: MESH Multi-Factory Flow (Multi-Site Owner)



### Three MESH highlights:

1. **Data sovereignty:** Each factory's data physically stays on-premise
2. **Offline-capable:** Factory keeps working even when HQ disconnects
3. **Infinite scale:** Adding the Nth outsource partner takes 50 lines of config

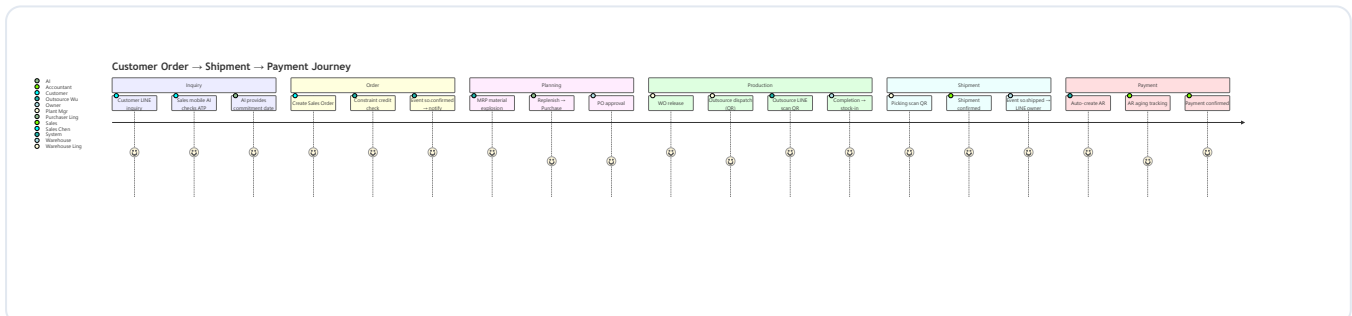
## View 5: Permission Check Flow (Security / IT)



Five layers of permission gates:

1. **JWT verification:** Is the token valid
2. **UserContext load:** Who are you, what role
3. **require\_permission:** Can you do this action
4. **apply\_row\_filter:** Which rows can you see (Sales A can't see Sales B's customers)
5. **Audit Trail:** Record who did what when

## View 6: Typical Business Lifecycle (Inquiry → Payment)



Events triggered at each stage:

| Stage      | Event               | Auto Action                           |
|------------|---------------------|---------------------------------------|
| Inquiry    | chat.completed      | DecisionLog records AI recommendation |
| Order      | so.confirmed        | Notify production_manager             |
| Planning   | mrp.generated       | Push to purchaser                     |
| Purchase   | po.approved         | Notify supplier + warehouse           |
| Production | wo.released         | Push to plant_manager + operators     |
| Outsource  | outsource.completed | Push to warehouse + accounting        |
| Shipment   | so.shipped          | LINE push to owner, auto-create AR    |
| Payment    | payment.received    | Notify sales + accounting             |

## Tech / Business Dual-Track Summary

| Dimension        | Tech View                       | Business View                         |
|------------------|---------------------------------|---------------------------------------|
| L1 Client        | React/Expo/HTML+SSE             | Owner / Sales / Plant Mgr / Outsource |
| L2 API Gateway   | FastAPI Middleware Stack        | "Identity verified before entry"      |
| L3a Multi-Agent  | IntentClassifier + Tool Calling | "AI auto-finds the right function"    |
| L3b Event Engine | EventBus + Constraint Rules     | "Anomalies push automatically"        |
| L4 Domain        | 12 service modules              | "12 business domains integrated"      |
| L5 Data          | PostgreSQL + Row-Level          | "Data isolated by factory & user"     |
| L6 MESH          | WireGuard + Aggregate Query     | "Outsource data stays at outsource"   |

## Reading Order by Role

| Reader                                                                                           | Suggested Order                                     |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|  Owner          | View 1 (30s scan) → View 6 (business journey)       |
|  Sales          | View 6 → View 2 (one request lifecycle)             |
|  Developer      | View 2 → View 3 (Multi-Agent) → View 5 (Permission) |
|  IT/Security    | View 5 (Permission) → View 4 (MESH)                 |
|  Multi-Site Mgr | View 4 (MESH) → View 1 (overview)                   |

## Related Documents

-  [ARCHITECTURE\\_DIAGRAM.md](#) — Static 5-layer
-  [NETWORK\\_DEPLOYMENT\\_EN.md](#) — Network deployment
-  [PERMISSION\\_MODEL.md](#) — Permission model
-  [LLM\\_BENCHMARK\\_REPORT\\_EN.md](#) — LLM benchmark
-  [ARCHITECTURE\\_DECISIONS.md](#) — ADRs

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